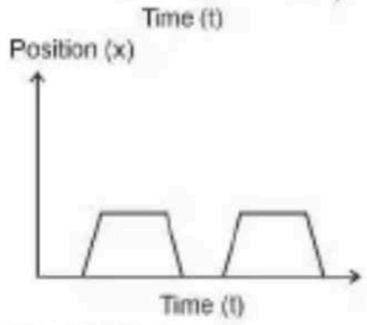
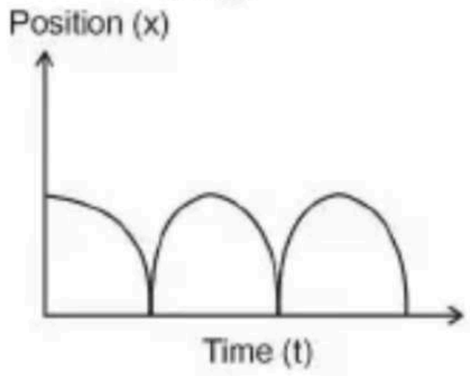
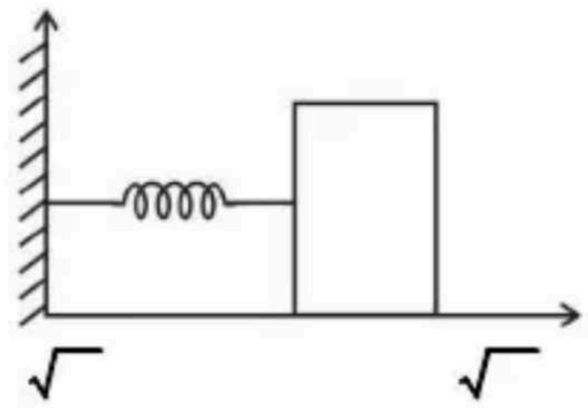
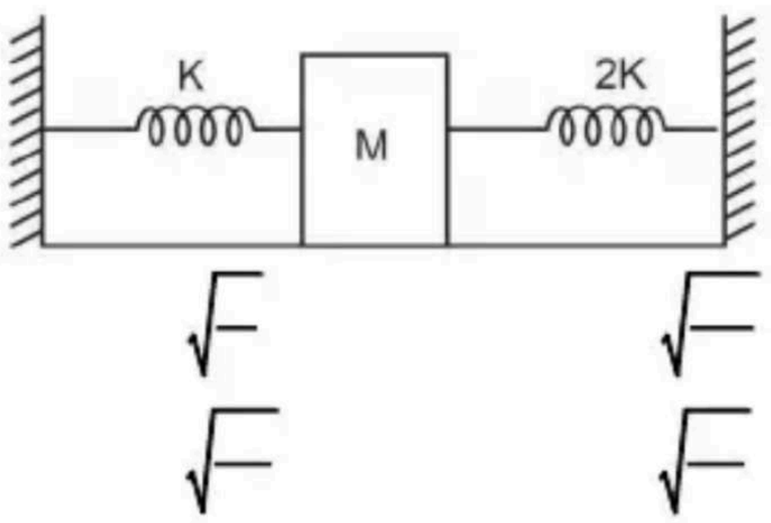
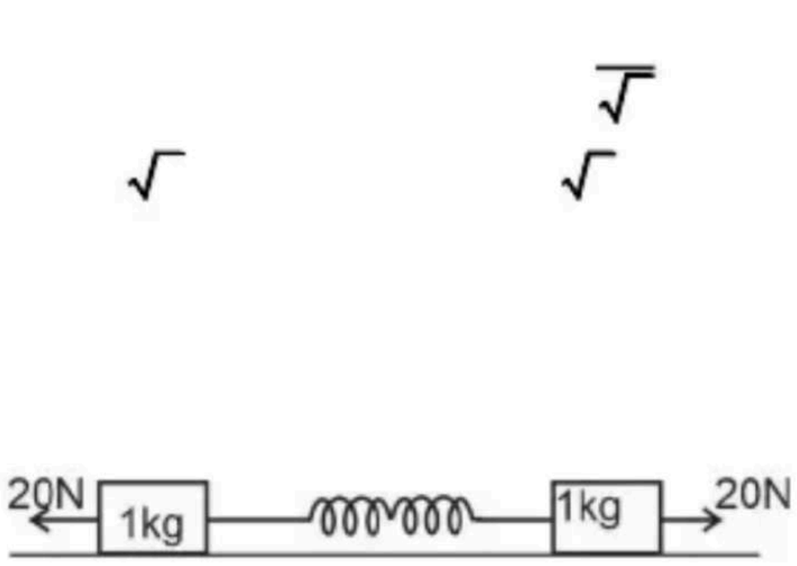

Position (x)
Time (t)

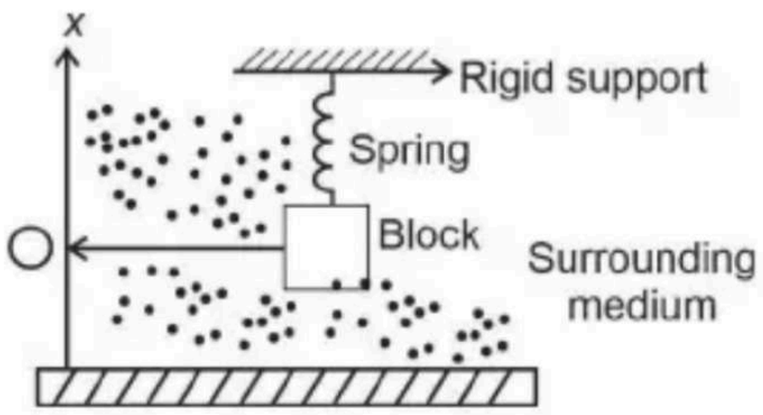

Position (x)
Time (t)


Position (x)
Time (t)




K
M
2K

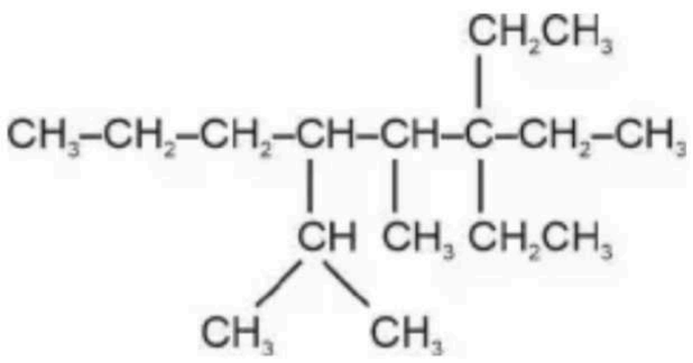

20N
1kg
1kg
20N

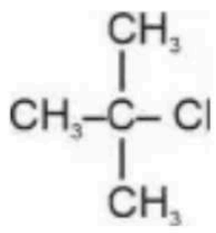


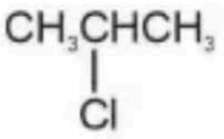
$$-$$

$$\sqrt{F}$$

–


CH2CH3
CH3–CH2–CH2–CH–CH–C–CH2–CH3
CH
CH3
CH2CH3
CH3
CH3


CH3
CH3–C– Cl
CH3

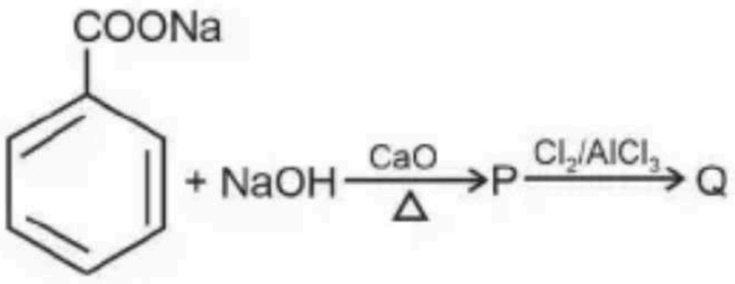
(iii) 

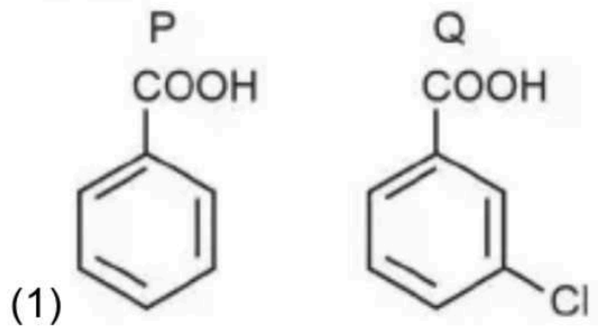
The correct order of the rate of β-elimination reaction with alc. KOH is

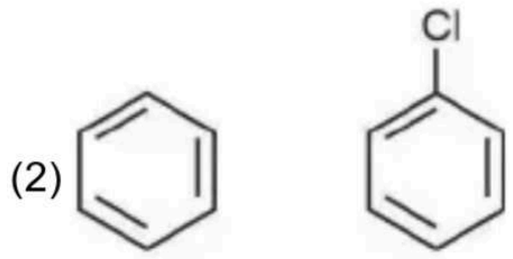
(1) (I) > (II) > (III) (2) (III) > (II) > (I)

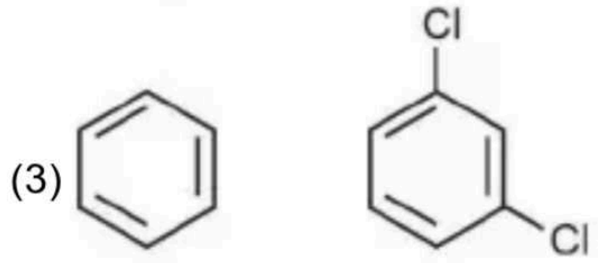
(3) (II) > (III) > (I) (4) (II) > (I) > (III)

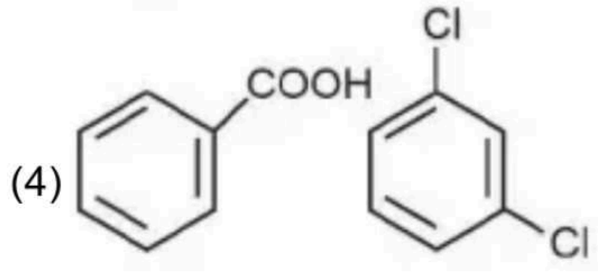
53. Identify the major products P and Q in the following sequence of reactions



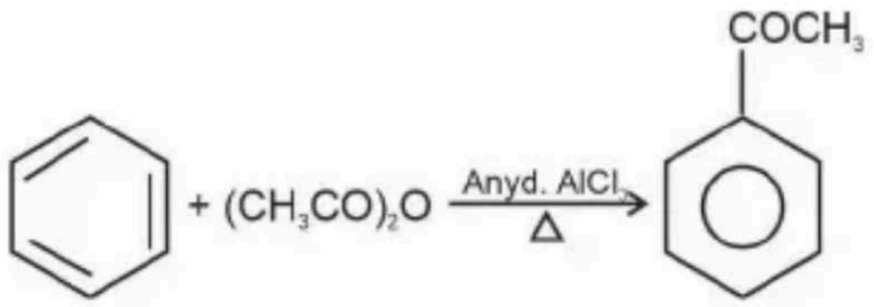
(1) 

(2) 

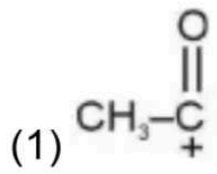
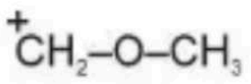
(3) 

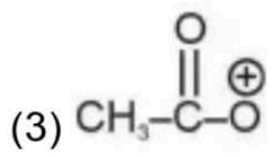
(4) 

54. In the reaction



The electrophile involved is

(1)  (2) 

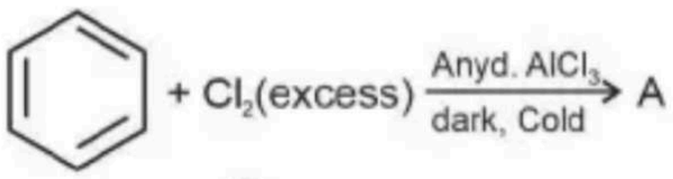
(3)  (4) CO_2

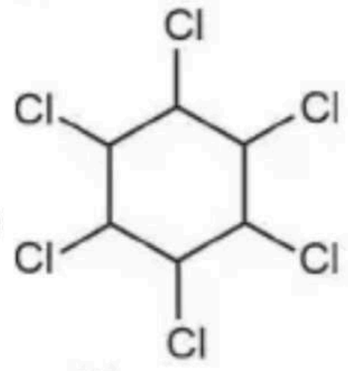
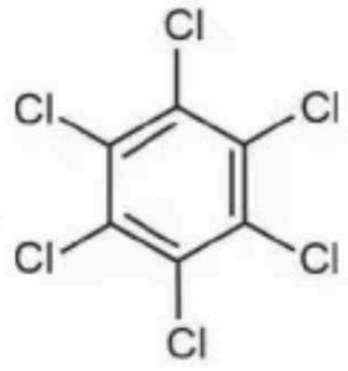
55. Number of sp^3 hybridised carbon in σ-complex or arenium ion is

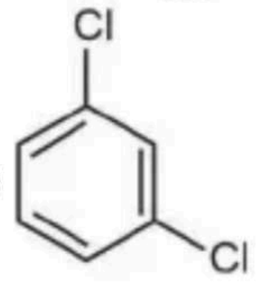
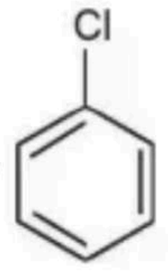
(1) 2 (2) 3

(3) 1 (4) 4

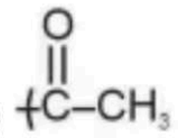
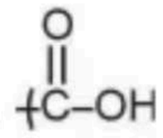
56. The correct structure of product A formed in the given reaction is

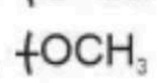


(1)  (2) 

(3)  (4) 

57. Which one of the following is an activating group?

(1)  (2) 

(3)  (4) 

58. Match the following columns and select the correct option

Column-I	Column-II
(a) Ionic solid	(i) HCl
(b) Metallic solid	(ii) Zns
(c) Network solid	(iii) Mg
(d) Molecular solid	(iv) SiC

	(a)	(b)	(c)	(d)
(1)	(I)	(II)	(III)	(IV)
(2)	(II)	(III)	(IV)	(I)
(3)	(IV)	(III)	(I)	(II)
(4)	(II)	(III)	(I)	(IV)

59. A trigonal crystal has

(1) a = b c; $\alpha = \beta = \gamma = 90°$

(2) a b = c; $\alpha = \beta = \gamma = 90°$

(3) a = b = c; $\alpha = \beta = \gamma$ 90°

(4) a b c; $\alpha = \beta = \gamma = 90°$

60. Atoms of element B form hcp lattice and those of the element A occupy – rd of tetrahedral voids. Then the formula of the compound formed by A and B is

(1) A_3B (2) AB_3

(3) A_3B_2 (4) A_2B_3

61. Volume occupied by all atoms present in a FCC unit cell of metal is (r = atomic radius)

(1) — (2) —

(3) — (4) –

62. Which of the following metal oxides behaves like metals

(1) CaO (2) CrO_2

(3) HgO (4) MgO

63. Which is the incorrect statement?

(1) Conductivities of semiconductors varies between 10^{-6} to 10^{4} ohm $^{1}m^{-1}$

(2) Diode is a combination of n-type and p-type semiconductors and is used as a rectifier

(3) Excess of potassium makes KCl crystals yellow

(4) Conductivity of insulator ranges between 10^{-20} to 10^{-10} ohm $^{1}m^{-1}$

64. Ethylene dibromide on reaction with Zinc metal gives

(1) CH CH

(2) $CH_2 = CH_2$

(3) CH_3 CH_2 Br

(4) CH_3 CH=CH CH_3

65. The edge length of body centred cubic unit cell is 300 pm. If the density of the metal is 5 g/cm^3 then number of atoms present in 200 g of the metal is

(1) 1.8×10^{23} atoms (2) 2.96×10^{24} atoms

(3) 6.022×10^{23} atoms (4) 12.04×10^{24} atoms

66. A compound forms hexagonal closed packed structure. If 1.5 mole of the compound is given then number of total voids will be

(1) 9.033×10^{23} (2) 6.022×10^{23}

(3) 27.09×10^{23} (4) 18.066×10^{23}

67. If A forms FCC structure then the coordination number of A will be

(1) 6 (2) 8

(3) 4 (4) 12

68. If 30% (mass/mass) aqueous solution of KI has density 1.02 g/mL then molarity of the solution will be (Atomic mass of K = 39 u and I = 127 u)

(1) 2.8 M (2) 1.4 M

(3) 2.5 M (4) 1.8 M

69. Correct order of solubility of gases in water at 298 K is

(1) Argon > Formaldehyde > Methane > Vinyl chloride

(2) Formaldehyde > Methane > Vinyl chloride > Argon

(3) Vinyl chloride > Formaldehyde > Methane > Argon

(4) Formaldehyde > Vinyl Chloride > Methane > Argon

70. If 62 g of ethylene glycol is mixed with 500 g water then freezing point of the solution will be (K_f of water = 1.86 K kg mol ^{1})

(1) 269.28 K (2) 270.95 K

(3) 273 K (4) 373 K

71. The boiling point of benzene is 353.23 K. When 2 g of non-volatile solute was dissolved in 100 g of benzene, the boiling point is raised to 355 K. Then molar mass of the solute is (K_b for benzene = 2.53 K kg mol ^{1})

(1) 28.58 g mol ^{1} (2) 58 g mol ^{1}

(3) 30 g mol ^{1} (4) 52.45 g mol ^{1}

72. Which among the following is not a component of photochemical smog?

(1) Ozone (2) Formaldehyde

(3) SO_2 (4) Acrolein

73. Three aqueous solution of $MgSO_4$ labelled as x, y and z with concentration 0.1M, 0.01M and

Hoff factor for these solutions will be in the order of

(1) $i_x > i_y > i_z$ (2) $i_z > i_y > i_x$

(3) $i_z > i_x > i_y$ (4) $i_y > i_x > i_z$

74. What amount of K_2SO_4 (i = 2.5) is dissolved in 3 L of water so that its osmotic pressure becomes 0.75 atm at 300 K?

(1) 6.4 g (2) 3.4 g

(3) 4 g (4) 8 g

75. When 23.9 g of $CHCl_3$ mixed with 42.5 g of CH_2Cl_2 at 298 K then vapour pressure of the solution becomes
= 200 mm Hg, = 400 mm Hg)

(1) 350 mm Hg (2) 250.5 mm Hg

(3) 342.8 mm Hg (4) 300 mm Hg

76. Order of depression in freezing point of solution when same concentration of acetic acid, trifluoro acetic acid and trichloro acetic acid is taken

(1) $CCl_3COOH < CF_3COOH < CH_3COOH$

(2) $CH_3COOH < CF_3COOH < CCl_3COOH$

(3) $CCl_3COOH < CH_3COOH < CF_3COOH$

(4) $CH_3COOH < CCl_3COOH < CF_3COOH$

77. Solution of azeotropic nitric acid mixture contains

(1) 32 % HNO_3 and 68 % H_2O by mass

(2) 68 % HNO_3 and 32 % H_2O by mass

(3) 95 % HNO_3 and 5 % H_2O by mass

(4) 5% HNO_3 and 95 % H_2O by mass

78. Which option is valid for an ideal solution?

(1) $\Delta G_{mix} > 0, \Delta H_{mix} = 0, \Delta V_{mix} = 0$

(2) $\Delta G_{mix} = 0, \Delta H_{mix} = 0, \Delta V_{mix} = 0$

(3) $\Delta G_{mix} < 0, \Delta H_{mix} = 0, \Delta V_{mix} = 0$

(4) $\Delta G_{mix} < 0, \Delta H_{mix}$ 0, ΔV_{mix} 0

79. Classical smog is a mixture of

(1) Fog and smoke only

(2) Smoke + Fog + SO_2

(3) Fog + SO_2 only

(4) $H_2O(g)$ + SO_2 only

80. Human excreta contains bacteria such as Escherichia coli and streptococcus faecalis, which cause

(1) Urinary diseases

(2) Respiratory diseases

(3) Gastro intestinal diseases

(4) Asthma

81. Polychlorinated biphenyls (PCBs) are

I Carcinogenic in nature

II Non-carcinogenic in nature

III Used as cleansing solvent

Select the correct option

(1) I and II (2) I and III

(3) II and III (4) I, II and III

82. The prescribed upper limit concentration of cadmium in drinking water is

(1) 0.005 ppm (2) 0.05 ppm

(3) 5 ppm (4) 2 ppm

83. Which tropic levels has highest concentration of DDT deposition?

(1) Larger fish (2) Small fish

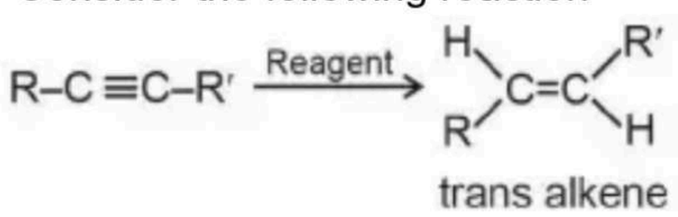
(3) Human being (4) Aquatic plant

84. Ferromagnetic substance among the following is

(1) MnO (2) Fe_3O_4

(3) Ni (4) $ZnFe_2O_4$

85. Consider the following reaction



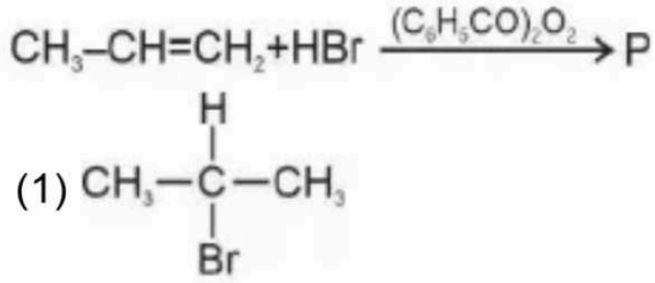
trans alkene

The suitable reagent is/are

(1) H_2/Pd/C (2) Na/liq. NH_3

(3) H_2/Ni (4) Both (1) and (2)

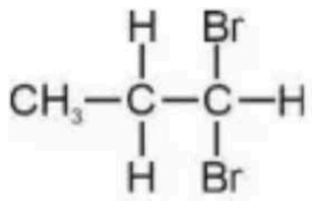
86. The major product (P) of the following reaction is



(1)

(2) Br CH_2 CH_2 CH_2 Br

(3) CH_3 CH_2 CH Br

(4) 

87. Which one is the correct order of acidicity?

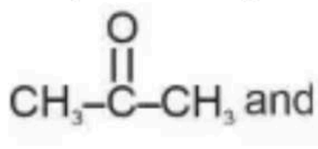
(1)

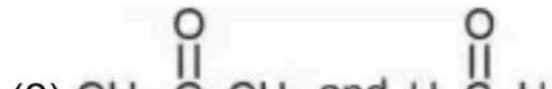
(2)

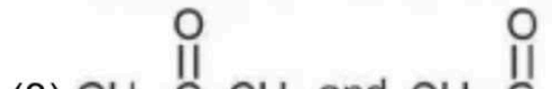
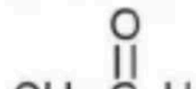
(3)

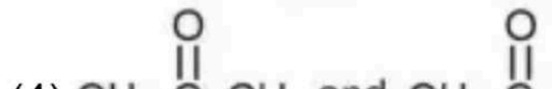
(4)

88. When 2-methyl propene undergoes reductive ozonolysis then products will be

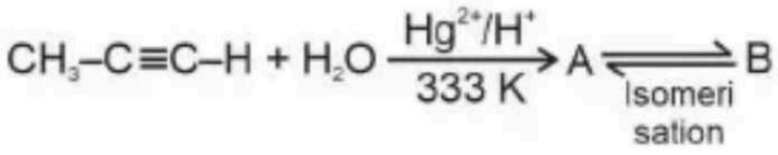
(1)  and CH_3COOH

(2)  and

(3)  and 

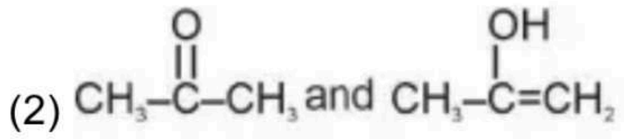
(4)  and 

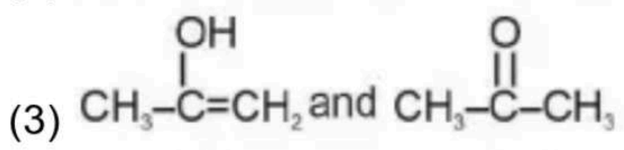
89. In the following reaction

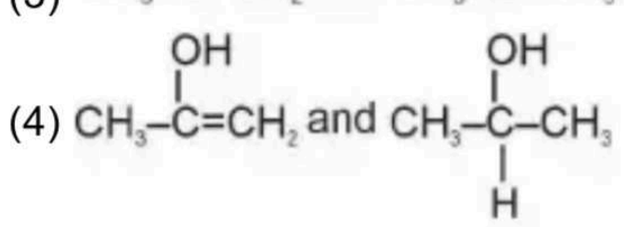


Here, A and B respectively are

(1) 

(2) 

(3) 

(4) 

90. Which solution shows negative deviation from ideal behaviour?

(1) Phenol + aniline (2) Ethanol + acetone

(3) CS_2 + Acetone (4) Both (2) and (3)

Botany

91. The degree by which progeny differ from their parents is called

(1) Inheritance (2) Variation

(3) Hybridisation (4) Pollination

92. Gregor Mendel, Conducted hybridisation experiments on garden pea during the year

(1) 1856-1863 (2) 1768-1775

(3) 1907-1914 (4) 1707-1714

93. Which of the given traits w.r.t. pea plant express itself in both homozygous as well as heterozygous state?

(A) Green pod (B) Axial flower

(C) Constricted pod (D) White flower

The **correct** ones are:

(1) (A) and (C) (2) (A) and (B)

(3) (B) and (D) (4) (C) and (D)

94. Read the statements (A-B) and select the **correct** option.

Statement (A): A true breeding line is one that, having undergone continuous self-pollination

Statement (B): Mendel proposed the laws of inheritance in living organisms.

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both statements (A) and (B) are correct

(4) Both statements (A) and (B) are incorrect

95. How many true-breeding pea plant varieties were selected by Mendel during his experiments of hybridisation?

(1) 7 (2) 14

(3) 12 (4) 6

96.

(1) TT TT (2) Tt Tt

(3) tt tt (4) Tt tt

97. The genotypic and phenotypic ratios respectively, obtained in F_2 generation during monohybrid cross, conducted by Mendel are

(1) 1:2:1 and 3:1 (2) 3:1 and 1:2:1

(3) 1:2:1 and 1:1 (4) 3:1 and 1:1

98. Law of dominance is used to best explain that

(1) There is blending of alleles

(2) There is recovery of both the characters as such in F_2-generation.

(3) The expression of only one of the parental characters in a monohybrid cross in F_1 and expression of both in F_2.

(4) Occurrence of alleles on same chromosome.

99. The F_1 phenotype did not resemble either of the two parents and was in between the two. This statement correctly explains which of the given inheritance pattern?

(1) Incomplete dominance

(2) Co-dominance

(3) Multiple allelism

(4) Pleiotropy

100. AB blood group is an example of

(1) Incomplete dominance

(2) Co-dominance

(3) Pleiotropy

(4) Polygenic inheritance

101. A gamete carries only one factor of a character. The statement is true for which of the given law of inheritance?

(1) Law of independent assortment

(2) Law of dominance

(3) Law of segregation

(4) Co-dominance

102. When a cross was made between red flowered and pink flowered , what will be the percentage of white flowered plants in progeny?

(1) 50% (2) 25%

(3) 100% (4) 0%

103.
getting plants which are homozygous for both seed shape and seed colour is

(1) 1/8 (2) 1/4

(3) 3/4 (4) 1/16

104. Chromosomal theory of inheritance was proposed by

(1) Sutton and Boveri (2) T.H. Morgan

(3) G.J. Mendel (4) Hugo de Vries

105. Males and females are heterogametic in (X) and

(Y) respectively.

Select the **correct** option of X and Y.

	(X)	(Y)
(1)		Birds
(2)	Birds	
(3)	Human being	Grasshopper
(4)	Grasshopper	Human being

106. Morgan selected as experimental material because of all of the given reasons, **except**

(1) could be grown on simple synthetic medium in the laboratory

(2) completes its life cycle in about two weeks.

(3) grows on fruits.

(4) has many types of hereditary variations that can be seen with power microscope.

107. Select the **incorrect** match

(1) ◇ - Sex unspecified

(2) ■ - Affected female

(3) □—○ - Mating

(4) □═○ - Mating between relatives (consanguineous mating)

108. Which of the given is autosomal dominant disorder?

(1) Myotonic dystrophy

(2) Sickle-cell anaemia

(3) Cystic fibrosis

(4) Colour blindness

109. Select the **incorrect** statement w.r.t.

(1) Is an autosomal recessive disorder.

(2) Shows its transmission from unaffected carries female to some of the male progeny.

(3) In this disease, a single protein that is a part of the Cascade of proteins involved in clotting of blood is affected.

(4) The possibility of female becoming a haemophilic is extremely rare.

110. In phenylketonuria, the affected individual lacks an enzyme that converts the amino acid A into B.

	A	B
(1)	Phenylalanine	Tyrosine
(2)	Glutamic acid	Valine
(3)	Tyrosine	Phenylalanine
(4)	Valine	Glutamic acid

111.
copy of

(1) Chromosome 11 (2) Chromosome 12

(3) Chromosome 21 (4) Chromosome 6

112. Failure of cytokinesis after teleophase stage of cell division results in

(1) Polyploidy (2) Aneuploidy

(3) (4) Haploidy

113. Which of the given genetic disorders is caused due to presence of an additional copy of X-chromosome?

(1)

(2)

(3) PKU

(4)

114. Karyotype of an individual who is inflicted with

(1) 45 + XO (2) 45 + XX

(3) 44 + XO (4) 45 + XXY

115. Match the Column-I and Column-II w.r.t. chromosomal disorder and select the **correct** option

Column-I	**Column-II**
A	(i) Gynaecomastia
B	(ii) Short statured with small round head, furrowed tongue and partially open mouth
C	(iii) Sterile female with underdeveloped feminine characters

	A	B	C
(1)	(ii)	(iii)	(i)
(2)	(i)	(ii)	(iii)
(3)	(iii)	(ii)	(i)
(4)	(i)	(iii)	(ii)

116. Frame-shift mutation occurs due to

(1) Insertion (2) Deletion

(3) Transversion (4) Both (1) and (2)

117. The physical association of genes on chromosome is called

(1) Linkage (2) Recombination

(3) Pleiotropy (4) Multiple allelism

118. Calculate the probability of a pea plant of being dwarf in F_1 when homozygous tall and dwarf plants are crossed together?

(1) 1/4 (2) 1/2

(3) 0 (4) 1

119. Amongst the following, which one has maximum numbers of nucleotides in its genome?

(1) $\phi \times 174$ bacteriophage (2)

(3) Human (4) Lambda phage

120. A nucleotide consists of all of the given components, **except**

(1) A nitrogenous base (2) Pentose sugar

(3) Phosphate (4) Hexose sugar

121. Which of the given nitrogenous base is found in RNA but not in DNA?

(1) Adenine (2) Uracil

(3) Cytosine (4) Guanine

122. The another chemical name of thymine is

(1) 5-methyl uracil (2) 7-methyl guanine

(3) 3-methyl uracil (4) 5-methyl guanosine

123. Euchromatin

(1) Is transcriptionally active

(2) Is transcriptionally inactive

(3) Stains dark

(4) Is more densely packed

124. Select the **incorrect** statement w.r.t. salient features of the Double-helix structure of DNA.

(1) Is made up of two polynucleotide chains

(2) The two chains have anti-parallel polarity

(3) The bases in two strands are paired through hydrogen bond

(4) Adenine forms three hydrogen bond with thymine

125. The pitch of the DNA helix is

(1) 2.2 nm (2) 2.4 nm

(3) 3.4 nm (4) 1.4 nm

126. According to central Dogma in molecular biology, the genetic information flows from

(1) DNA $\rightarrow$ RNA $\rightarrow$ Protein

(2) DNA $\rightarrow$ Protein $\rightarrow$ RNA

(3) RNA $\rightarrow$ DNA $\rightarrow$ Protein

(4) Protein $\rightarrow$ RNA $\rightarrow$ DNA

127. Central Dogma in molecular biology was proposed by

(1) Francis Crick (2) J. Watson

(3) T.H. Morgan (4) A.H. Sturtevant

128. Histones are rich in the basic amino acid residues A and B .

(1) A – Histidine, B – Arginine

(2) A – Lysine, B – Arginine

(3) A – Lysine, B – Histidine

(4) A – Valine, B – Glycine

129. A typical nucleosome contains how much nucleotides of DNA helix?

(1) 100 bases (2) 100 bp

(3) 200 bases (4) 200 bp

130. Select the **incorrect** interpretation w.r.t. transforming principle

(1) S strain → Inject into mice → Mice die

(2) R strain → Inject into mice → Mice live

(3) S strain (heat killed) → Inject into mice → Mice live

(4) S strain (heat killed) + R strain (live) → Inject into mice → Mice survived

131. The principle enzyme involved in DNA replication is

(1) DNA dependent DNA polymerase

(2) DNA dependent RNA polymerase

(3) RNA primase

(4) DNA Ligase

132. In eukaryotes, the replication of DNA takes place at which of the given phase of cell cycle events?

(1) S-phase (2) G_1 phase

(3) G_2 phase (4) G_0 phase

133. Read the statement (A-C) stating them true (T) or false (F) and select the correct option.

A The DNA polymerase on their own can initiate the process of replication.

B The process of copying genetic information from one strand of DNA into RNA is termed as 'transcription'.

C Replication is discontinuous on the template with polarity 5'→3'.

	A	B	C
(1)	T	T	F
(2)	F	T	F
(3)	T	T	T
(4)	F	T	T

134. In a double stranded DNA, the proportion of thymine was found to be 26%, then calculate the amount of guanine?

(1) 26% (2) 24%

(3) 37% (4) 52%

135. In Hershey-chase experiment, the DNA of T_2-phage was made radioactive by using

(1) ^{32}P (2) ^{35}S

(3) ^{32}S (4) ^{35}P

Zoology

136. Read following statements **A** and **B** and choose the **correct** option.

(A) The period from birth to natural death of an organism represents its life span

(B) No individual is immortal, except single-celled organisms

(1) Both the statements A and B are **correct**

(2) Both the statements A and B are **incorrect**

(3) Only statement A is **incorrect**

(4) Only statement B is **incorrect**

137. All of the following organisms have life span in years **except**

(1) Crow (2) Butterfly

(3) Tortoise (4) Parrot

138. Select **incorrect** statement w.r.t. reproduction.

(1) Reproduction enables the continuity of the species, generation after generation

(2) Sexually reproducing organism gives rise to offspring similar to itself

(3) In asexual reproduction, offspring is produced by a single parent with or without formation of gamete

(4) Fusion of male and female gametes involves in both sexual and asexual reproduction

139. Clones are **not**

(1) Product of asexual reproduction

(2) Morphologically similar individual

(3) Genetically similar individuals

(4) Product of sexual reproduction

140. Cell division is itself a mode of reproduction in

(1) (2)
(3) (4)

141. Pseudopodiospores of are liberated in the surrounding medium
(1) During unfavourable condition
(2) For encystation
(3) During favourable condition
(4) To grow into a single

142. In comparison to asexual reproduction, sexual reproduction is
(1) An elaborate and complex process
(2) Fast process
(3) Not significant for evolution
(4) Generally uniparental

143. How many organisms in following given box represent cyclical changes during their reproduction phase?

Sheep, Rat, Dear, Tiger, Human, Monkey, Ape

(1) 4 (2) 3
(3) 5 (4) 7

144. All of the following are events in sexual reproduction in humans **except**
(1) Senescence after menopause
(2) Gametogenesis in females
(3) Formation of zygote
(4) Embryogenesis

145. Read following statements carefully and choose the option with all **correct** statements.
(a) Fusion of gametes is perhaps the most vital events of sexual reproduction
(b) Zygote is the vital link that ensures continuity of species between organisms of one generation and the next
(c) Organisms exhibiting external fertilization release a large number of sperms in female genital tract
(d) Cell differentiation helps the cells to increase in their number
(1) a and b (2) b and c
(3) c and d (4) a and d

146. The reproductive events in humans which leads to transfer of sperms into the female genital tract is called
(1) Fertilization (2) Capacitation
(3) Insemination (4) Spermiation

147. Select the **incorrect** match.
(1) Gametogenesis – Formation of germ cells
(2) Implantation – Attachment of blastocyst to the uterine wall
(3) Gestation – Embryonic development
(4) Parturition – Delivery of the baby

148. Each testis has about **A** compartments called testicular lobules and each lobule contains **B** seminiferous tubules.
Choose the option which **correctly** represent **A** and **B**

S. No	A	B
1	150	4 – 6
2	250	6 – 8
3	150	1 – 3
4	250	1 – 3

149. Which of the following structure is **not** present in interstitial space of testis?
(1) Leydig cells
(2) Immunologically competent cells
(3) Sertoli cells
(4) Small blood vessels

150. Choose the intra-abdominal male sex accessory duct among following given structures.
(1) Rete testis
(2) Vasa efferentia
(3) Epididymis
(4) Vas deferens

151. Epididymis is located along which surface of each testis?
(1) Anterior (2) Posterior
(3) Medial (4) Lateral

152. The structure(s) which is/are responsible for storage and transport of sperms from the testis to the outside through urethra is/are
(1) Seminal vesicle
(2) Prostate gland
(3) Male sex accessory ducts
(4) Penis

153. Select the option with all **incorrect** statements w.r.t. male reproductive system.
(a) Muscle contraction causes erection of penis which facilitates insemination
(b) All accessory sex glands in male are paired
(c) Seminal plasma is rich in glucose, calcium and certain enzymes

(d) The secretions of bulbourethral glands also help in the lubrication of penis

(1) Only a and b
(2) a, b and c
(3) a, b, c and d
(4) b, c and d

154. External genitalia in human females excludes
(1) Hymen
(2) Clitoris
(3) Vagina
(4) Labia minora

155. Each ovary is connected by ligaments to the
(1) Wall of uterus only
(2) Pelvic wall only
(3) Pelvic wall and uterus
(4) Wall of vagina

156. Which of the following structures is **not** included in female accessory ducts?
(1) Oviducts
(2) Uterus
(3) Vagina
(4) Lactiferous ducts

157. Birth canal is formed
(1) Only by cervix
(2) Only by vagina
(3) Uterus along with cervical canal
(4) Cervical canal along with vagina

158. Which of the following layer of uterus undergoes cyclical changes during menstrual cycle?
(1) Perimetrium
(2) Myometrium
(3) Endometrium
(4) Perimetrium and endometrium

159. Cluster of cells present in mammary lobes responsible for milk secretion are called
(1) Alveoli
(2) Mammary tubules
(3) Mammary ducts
(4) Mammary ampulla

160. Choose the option which fills the blank **correct**ly to complete the analogy
Male germ cell: Spermatogonia: : gamete mother cells: _____
(1) Oogonia
(2) Polar bodies
(3) Primary oocyte
(4) Secondary oocyte

161. Choose the **odd** one w.r.t. ploidy.
(1) Primary spermatocyte
(2) Spermatogonia
(3) Primary oocyte
(4) First polar body

162. Which of the following is **correct** sequence of plasma concentration of various hormones to start spermatogenesis in humans at the age of puberty?
(1) Increase in FSH→ increase in GnRH→ decrease in androgens→ spermatogenesis
(2) Increase in gonadotropins →decrease in GnRH →increase in testosterone→ stimulates spermatogenesis
(3) Increase in GnRH→ increase in gonadotropins→ increase in androgens → stimulates spermatogenesis
(4) Increase in androgens → decrease in GnRH→ decrease gonadotropins→ stimulates spermatogenesis

163. FSH acts on Sertoli cells and stimulates secretion of some factors to help in process of
(1) Spermiogenesis
(2) Formation of spermatids
(3) Insemination
(4) Release of semen

164. In a normal adult human, 180 million sperms are found to have normal shape and size per ejaculate. Calculate least number of sperms in ejaculate which must show vigorous motility for normal fertility.
(1) 72 lakhs (2) 72 million
(3) 40 million (4) 60 lakh

165. Primary oocyte within the A grows in size and completes its B division. Choose the option which represents A and B **correctly**

	A	B
1.	Tertiary follicle	1st mitotic
2.	Graafian follicle	2nd meiotic
3.	Tertiary follicle	1st meiotic
4.	Graafian follicle	2nd meiotic

166. Which of the following is **mismatch** w.r.t. menstrual cycle?
(1) *Menarch* – First menstruation
(2) Menstrual cycle – Cycle of events in between end of two

successive menstruations

(3) Menstrual phase – Occurrence of menstrual flow

(4) Lack of menstruation – Can be indicative of pregnancy

167. Which of the following set of hormones represents their **correct** plasma concentrations in a fertile female at the time of ovulation?
(1) Peak level of FSH and LH
(2) Peak level oestrogens and progesterone
(3) Peak level of FSH and estrogens and minimum level of LH and progesterone
(4) Peak level of LH and oestrogens, moderate level of FSH and minimum level of progesterone

168. 2nd meiotic division during oogenesis is completed
(1) Before fertilization starts
(2) After fertilization
(3) During fertilization and before fusion of nuclei
(4) During fertilization and before formation of ootid

169. Sex of baby is decided at the time of fertilization depending upon
(1) Sex chromosome of females
(2) Only X chromosome of sperm
(3) Only Y chromosome of sperm
(4) Either X or Y chromosome of sperm

170. In humans, structural and functional unit between embryo and maternal body is formed by
(1) Fallopian tube
(2) Birth canal
(3) Chorionic villi and uterine tissue
(4) Inner cell mass

171. Select **incorrect** match.

1. Formation of embryo heart	–	After first trimester of pregnancy
2. Limbs and digits develops	–	After 2nd month of pregnancy
3. First movement of foetus	–	During the 5th month of pregnancy
4. Eye-lids separate	–	By the end of 6th month

172. Foetal ejection reflex triggers release of
(1) Oxytocin from foetal adenohypophysis
(2) Oxytocin from foetal neurohypophysis
(3) Prolactin from maternal pituitary
(4) Oxytocin from maternal pituitary

173. RCH stands for
(1) Routine check-up of health
(2) Reproduction cum hygiene
(3) Reproduction and community health
(4) Reproductive and child health care

174. The procedure in which fluid around foetus is taken for detection of foetal cells and dissolved substances is
(1) PCR
(2) ELISA
(3) Amniocentesis
(4) Ultrasonography

175. Which of the following disease is completely curable if detected early and treated properly?
(1) Syphilis (2) AIDS
(3) Hepatitis-B (4) Genital herpes

176. Inability to conceive or produce children even after 2 years of unprotected sexual cohabitation is called
(1) Infertility (2) Fertility
(3) MTP (4) PID

177. A test tube baby differs from normally produced baby in
(1) Parturition
(2) Gestation period
(3) *In vitro* fertilization
(4) Number of chromosome

178. All of the following are natural method of contraception **except**
(1) Periodic abstinence
(2) Lactational amenorrhoea
(3) Barrier methods
(4) Coitus interruptus

179. *Progestasert* and LNG-20 are
(1) Injectable estrogen
(2) Hormone releasing IUDs
(3) Copper releasing IUDs
(4) Non-medicated IUDs

180. *In Vitro* fertilization occurs in
(1) ZIFT (2) GIFT
(3) IUI (4) AI

❑❑❑

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3.	(3)	39.	(3)	75.	(3)	111.	(3)	147.	(1)
4.	(2)	40.	(3)	76.	(4)	112.	(1)	148.	(4)
5.	(1)	41.	(3)	77.	(2)	113.	(4)	149.	(3)
6.	(2)	42.	(3)	78.	(3)	114.	(3)	150.	(4)
7.	(4)	43.	(1)	79.	(2)	115.	(1)	151.	(2)
8.	(3)	44.	(2)	80.	(3)	116.	(4)	152.	(3)
9.	(3)	45.	(2)	81.	(2)	117.	(1)	153.	(2)
10.	(3)	46.	(4)	82.	(1)	118.	(3)	154.	(3)
11.	(4)	47.	(2)	83.	(3)	119.	(3)	155.	(3)
12.	(3)	48.	(4)	84.	(3)	120.	(4)	156.	(4)
13.	(1)	49.	(1)	85.	(2)	121.	(2)	157.	(4)
14.	(4)	50.	(3)	86.	(3)	122.	(1)	158.	(3)
15.	(4)	51.	(2)	87.	(2)	123.	(1)	159.	(1)
16.	(3)	52.	(3)	88.	(2)	124.	(4)	160.	(1)
17.	(4)	53.	(2)	89.	(3)	125.	(3)	161.	(4)
18.	(2)	54.	(1)	90.	(1)	126.	(1)	162.	(3)
19.	(3)	55.	(3)	91.	(2)	127.	(1)	163.	(1)
20.	(4)	56.	(2)	92.	(1)	128.	(2)	164.	(2)
21.	(3)	57.	(4)	93.	(2)	129.	(4)	165.	(3)
22.	(1)	58.	(2)	94.	(3)	130.	(4)	166.	(2)
23.	(1)	59.	(3)	95.	(2)	131.	(1)	167.	(1)
24.	(1)	60.	(4)	96.	(4)	132.	(1)	168.	(3)
25.	(4)	61.	(3)	97.	(1)	133.	(4)	169.	(4)
26.	(3)	62.	(2)	98.	(3)	134.	(2)	170.	(3)
27.	(1)	63.	(3)	99.	(1)	135.	(1)	171.	(1)
28.	(4)	64.	(2)	100.	(2)	136.	(1)	172.	(4)
29.	(2)	65.	(2)	101.	(3)	137.	2)	173.	(4)
30.	(4)	66.	(3)	102.	(4)	138.	(4)	174.	(3)
31.	(1)	67.	(4)	103.	(2)	139.	(4)	175.	(1)
32.	(2)	68.	(4)	104.	(1)	140.	(1)	176.	(1)
33.	(4)	69.	(2)	105.	(1)	141.	(3)	177.	(3)
34.	(2)	70.	(1)	106.	(3)	142.	(1)	178.	(3)
35.	(1)	71.	(1)	107.	(2)	143.	(4)	179.	(2)
36.	(4)	72.	(3)	108.	(1)	144.	(1)	180.	(1)

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13. Answer (1)
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14. Answer (4)
NCERT Reference:
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15. Answer (4)
NCERT Reference:
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16. Answer (3)
NCERT Reference:
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17. Answer(4)
NCERT Reference:
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18. Answer(2)
NCERT Reference:
Physics XI(Part II) Page No.-352
19. Answer(3)
NCERT Reference:
Physics XI(Part II) Page No. -351
20. Answer (4)

NCERT Reference:

Physics XI(Part II) Page No. -360

21. Answer(3)

NCERT Reference:

Physics XI(Part II) Page No.-361

22. Answer(1)

NCERT Reference:

Physics XI(Part II) Page No-359

23. Answer (1)

NCERT Reference:

Physics XI(Part II) Page No. -366

24. Answer (1)

NCERT Reference:

Physics XI(Part II) Page No.-386

25. Answer (4)

NCERT Reference:

Physics XI(Part II) Page No.-386

26. Answer (3)

NCERT Reference:

Physics XI(Part II) Page No.386

27. Answer (1)

NCERT Reference:

Physics XI(Part II) Page No.-386

28. Answer (4)

NCERT Reference:

Physics XI(Part II) Page No.-368

29. Answer (2)

NCERT Reference:

Physics XI(Part II) Page No.-370

30. Answer (4)

NCERT Reference:

Physics XI(Part II) Page No.-387

31. Answer (1)

NCERT Reference:

Physics XI(Part II) Page No.-387

32. Answer (2)

NCERT Reference:

Physics XI(Part II) Page No.-387

33. Answer (4)

NCERT Reference:

Physics XI(Part II) Page No.-377

34. Answer (2)

NCERT Reference:

Physics XI(Part II) Page No.-380

35. Answer(1)

NCERT Reference:

Physics XI(Part II) Page No.-383

36. Answer(4)

NCERT Reference:

Physics XI(Part II) Page No.-387

37. Answer (4)

NCERT Reference:

Physics XI(Part II) Page No.-388

38. Answer(3)

NCERT Reference:

Physics XI(Part II) Page No.-387

39. Answer(3)

NCERT Reference:

Physics XI(Part II) Page No.-388

40. Answer(3)

NCERT Reference:

Physics XI(Part II) Page No.-387

41. Answer(3)

NCERT Reference:

Physics XI(Part II) Page No.-388

42. Answer(3)

NCERT Reference:

Physics XI(Part II) Page No.-389

43. Answer(1)

NCERT Reference:

Physics XI(Part II) Page No.-389

44. Answer(2)

NCERT Reference:

Physics XI(Part II) Page No.-390

45. Answer(2)

NCERT Reference:

Physics XI(Part II) Page No.-388

46. Answer (4)

NCERT Reference : Hydrocarbon, XI, Part-II, Page No.- 376

47. Answer (2)

NCERT Reference : Hydrocarbon, XI, Part-I, Page No.-377

48. Answer (4)

NCERT Reference : Hydrocarbon XI Part-I, Page No.-379

49. Answer (1)

NCERT Reference :Hydrocarbon, XI, Part-II, Page No.-380

50. Answer (3)

NCERT Reference : Hydrocarbon, XI, Part-II, Page No.-381

51. Answer (2)

NCERT Reference : Hydrocarbon, XI, Part-II, Page No.-391

52. Answer (3)

NCERT Reference : Hydrocarbon, XI Part-II, Page No.-388

53. Answer (2)

NCERT Reference : Hydrocarbon, XI, Part-II, Page No.-400

54. Answer (1)

NCERT Reference : Hydrocarbon, XI, part-II, Page No.- 401

55. Answer (3)

NCERT Reference : Hydrocarbon, XI Part-II, Page No.-401

56. Answer (2)

NCERT Reference : Hydrocarbon, XI, Part-II, Page No.401

57. Answer (4)

NCERT Reference : Hydrocarbon, XI, Part-II Page No.-403

58. Answer (2)

NCERT Reference : Solid state, XII, Part-I, Page No.-5

59. Answer (3)

NCERT Reference : Solid state, XII, Part-I, Page No.-10

60. Answer (4)

NCERT Reference : Solid state, XII part-I, page No.-18

61. Answer (3)

NCERT Reference : Solid state, XII, Part-I page No.-20

62. Answer (2)

NCERT Reference : Solid state, XII, Part-I, Page No.-29

63. Answer (3)

NCERT Reference : Solid state, XII, Part-I, Page No.-27 & 29

64. Answer (2)

NCERT Reference : Hydrocarbons, XI, Part-II, Page No.-388

65. Answer (2)

NCERT Reference : Solid state, XII, Part-I, Page No.-22

66. Answer (3)

NCERT Reference : Solid state, XII, Part-I, Page No.-23

67. Answer (4)

NCERT Reference : Solid state, XII, Part-I, Page No.-18

68. Answer (4)

NCERT Reference : Solution, XII, Part-I, Page No.-39

69. Answer (2)

NCERT Reference : Solution, XII, Part-I, Page No.-41

70. Answer (1)

NCERT Reference : Solution, XII, Part-I, Page No.-54

71. Answer (1)

NCERT Reference : Solution, XII, Part-I, Page No.-52

72. Answer (3)

NCERT Reference : Environmental chemistry, XI, Part-II, Page No.- 412

73. Answer (2)

NCERT Reference : Solution, XII, Part-I, Page No.-59

74. Answer (1)

NCERT Reference : Solution, XII, Part-I, Page No.-59

75. Answer (3)

NCERT Reference : Solution, XII, Part-I, Page No.-45

76. Answer (4)

NCERT Reference : Solution, XII, Part-I, Page No.-63

77. Answer (2)

NCERT Reference : Solution, XII, Part-I, Page No.-49

78. Answer (3)

NCERT Reference : Solution, XII, Part-I, Page No.-47

79. Answer (2)

NCERT Reference : Environmental chemistry, XI, Part-II, Page No.- 411

80. Answer (3)

NCERT Reference : Environmental chemistry, XI, Part-II, Page No.-414

81. Answer (2)

NCERT Reference : Environmental chemistry, XI, Part-II, Page No-415

82. Answer (1)

NCERT Reference : Environmental chemistry, XI Part-II, Page No.- 416

83. Answer (3)

NCERT Reference : Environmental chemistry, XI, Part-II, Page No.-417

84. Answer (3)

NCERT Reference : Solid state, XII, Part-II, Page No.-30

85. Answer (2)

NCERT Reference : Hydrocarbons, XI, Part-II, Page No.-387

86. Answer (3)

NCERT Reference : Hydrocarbons, XI, Part-II, Page No.-390

87. Answer (2)

NCERT Reference : Hydrocarbons, XI, Part-II, Page No.-394

88. Answer (2)

NCERT Reference : Hydrocarbon, XI, Part-II, Page No.-391

89. Answer (3)

NCERT Reference : Hydrocarbon, XI, Part-II, Page No.-395

90. Answer (1)

NCERT Reference : Solutions, XII, Part-I, Page No.-48

91. Answer (2)

NCERT Reference : (Biology-Class XII) Page No. 69

92. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 70

93. Answer (2)

NCERT Reference : (Biology-Class XII) Page No. 70

94. Answer (3)

NCERT Reference : (Biology-Class XII) Page No. 70

95. Answer (2)

NCERT Reference : (Biology-Class XII) Page No. 70

96. Answer (4)

NCERT Reference : (Biology-Class XII) Page No. 74

97. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 73

98. Answer (3)

NCERT Reference : (Biology-Class XII) Page No. 75

99. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 76

100. Answer (2)

NCERT Reference : (Biology-Class XII) Page No. 77

101. Answer (3)

NCERT Reference : (Biology-Class XII) Page No. 75

102. Answer (4)

NCERT Reference : (Biology-Class XII) Page No. 76

103. Answer (2)

NCERT Reference : (Biology-Class XII) Page No. 79

104. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 81

105. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 86

106. Answer (3)

NCERT Reference : (Biology-Class XII) Page No. 83

107. Answer (2)

NCERT Reference : (Biology-Class XII) Page No. 88

108. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 88

109. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 89

110. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 90

111. Answer (3)

NCERT Reference : (Biology-Class XII) Page No. 91

112. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 90-91

113. Answer (4)

NCERT Reference : (Biology-Class XII) Page No. 91

114. Answer (3)

NCERT Reference : (Biology-Class XII) Page No. 91

115. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 91

116. Answer (4)

NCERT Reference : (Biology-Class XII) Page No. 87

117. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 83

118. Answer (3)

NCERT Reference : (Biology-Class XII) Page No. 72

119. Answer (3)

NCERT Reference : (Biology-Class XII) Page No. 96

120. Answer (4)

NCERT Reference : (Biology-Class XII) Page No. 96

121. Answer (2)

NCERT Reference : (Biology-Class XII) Page No. 96

122. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 97

123. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 100

124. Answer (4)

NCERT Reference : (Biology-Class XII) Page No. 97

125. Answer (3)

NCERT Reference : (Biology-Class XII) Page No. 97

126. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 98

127. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 98

128. Answer (2)

NCERT Reference : (Biology-Class XII) Page No. 99

129. Answer (4)

NCERT Reference : (Biology-Class XII) Page No. 99

130. Answer (4)

NCERT Reference : (Biology-Class XII) Page No. 100

131. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 106

132. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 107

133. Answer (4)

NCERT Reference : (Biology-Class XII) Page No. 106-107

134. Answer (2)

NCERT Reference : (Biology-Class XII) Page No. 97

135. Answer (1)

NCERT Reference : (Biology-Class XII) Page No. 102

136. Answer (1)

NCERT Reference: Page No.-03

137. Answer(2)

NCERT Reference: Page No.-04

138. Answer (4)

NCERT Reference: Page No.-05

139. Answer (4)

NCERT Reference: Page No.-05

140. Answer(1)

NCERT Reference: Page No.-05

141. Answer (3)

NCERT Reference: Page No.-06

142. Answer (1)

NCERT Reference: Page No.-08

143. Answer (4)

NCERT Reference: Page No.-09

144. Answer (1)

NCERT Reference: Page No. 10

145. Answer (1)

NCERT Reference: Page No. -15

146. Answer (3)

NCERT Reference: Page No. -42

147. Answer (1)

NCERT Reference: Page No. -42

148. Answer (4)

NCERT Reference: Page No. -43

149. Answer (3)

NCERT Reference: Page No. -43

150. Answer (4)

NCERT Reference: Page No.-43

151. Answer (2)

NCERT Reference: Page No.-43

152. Answer(3)

NCERT Reference: Page No. -43

153. Answer(2)

NCERT Reference: Page No.-44

154. Answer(3)

NCERT Reference: Page No. -44

155. Answer (3)

NCERT Reference: Page No. -44

156. Answer(4)

NCERT Reference: Page No.-45

157. Answer(4)

NCERT Reference: Page No-46

158. Answer (3)

NCERT Reference: Page No. -46

159. Answer (1)

NCERT Reference: Page No.-47

160. Answer (1)

NCERT Reference: Page No.-48

161. Answer (4)

NCERT Reference: Page No.48

162. Answer (3)

NCERT Reference: Page No.-47

163. Answer (1)

NCERT Reference: Page No.-47

164. Answer (2)

NCERT Reference: Page No.-48

165. Answer (3)

NCERT Reference: Page No.-48

166. Answer (2)

NCERT Reference: Page No.-49,50

167. Answer (1)

NCERT Reference: Page No.-50

168. Answer (3)

NCERT Reference: Page No.-52

169. Answer (4)

NCERT Reference: Page No.-52

170. Answer(3)

NCERT Reference: Page No.-53

171. Answer(1)

NCERT Reference: Page No.-54

172. Answer (4)

NCERT Reference: Page No.-54

173. Answer(4)

NCERT Reference: Page No.-58

174. Answer(3)

NCERT Reference: Page No.-58

175. Answer(1)

NCERT Reference: Page No.-63

176. Answer(1)

NCERT Reference: Page No.-63

177. Answer(3)

NCERT Reference: Page No.-64

178. Answer(3)

NCERT Reference: Page No.-60

179. Answer(2)

NCERT Reference: Page No.-60

180. Answer(1)

NCERT Reference: Page No.-64

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